



NAME:

TEACHER:

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<<School Name>>

Level 1 Mathematics and Statistics

Practice Exam

91947

Demonstrate mathematical reasoning

Credits: Five

Achievement	Achievement with Merit	Achievement with Excellence
Demonstrate mathematical reasoning.	Demonstrate mathematical reasoning with relational thinking.	Demonstrate mathematical reasoning with extended abstract thinking.

You should attempt ALL the questions in this booklet.

If you need more room for any answer, use the extra space provided at the back of this booklet.

Check that this booklet has pages 2–8 in the correct order and that none of these pages is blank.

Do not write in the grey shaded columns on the right of each page. This area is for markers only.

YOU MUST HAND THIS BOOKLET TO THE SUPERVISOR AT THE END OF THE EXAMINATION.

Links made from Subject Learning Outcome for Assessments document

<https://ncea-live-3-storage-53q-assetstorage3bucket-2o21xte0r81u.s3.amazonaws.com/s3fs-public/2024-04/MS%20NCEA%20Level%201%20Subject%20Learning%20Outcomes.pdf?VersionId=QXvuELZpwFBw66s9JLnV2XFqE4crufa2>

QUESTION ONE

- (a) If $s = t \left(\frac{u+v}{2} \right)$, give the equation for u in terms of s , t and v .

rearranging, and using formulas (excluding the use of logarithms) – examples include: $d = \frac{1}{2}at^2$, $A = P(1 + nr)$, $t =$

$$2\pi \sqrt{\frac{L}{g}}$$

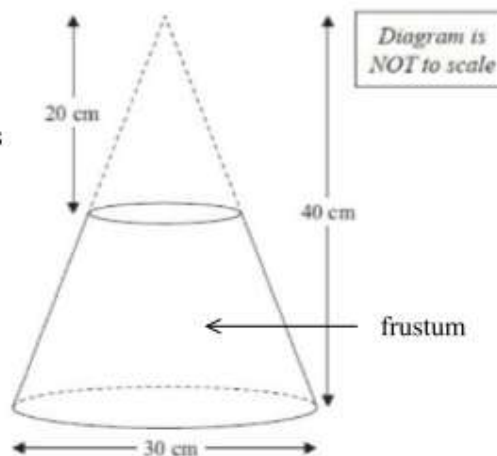
- (b) A frustum is made by removing a small cone from a similar large cone.

The height of the small cone is 20 cm.

The height of the large cone is 40 cm.

The diameter of the base of the large cone is 30 cm.

Work out the volume of the frustum.



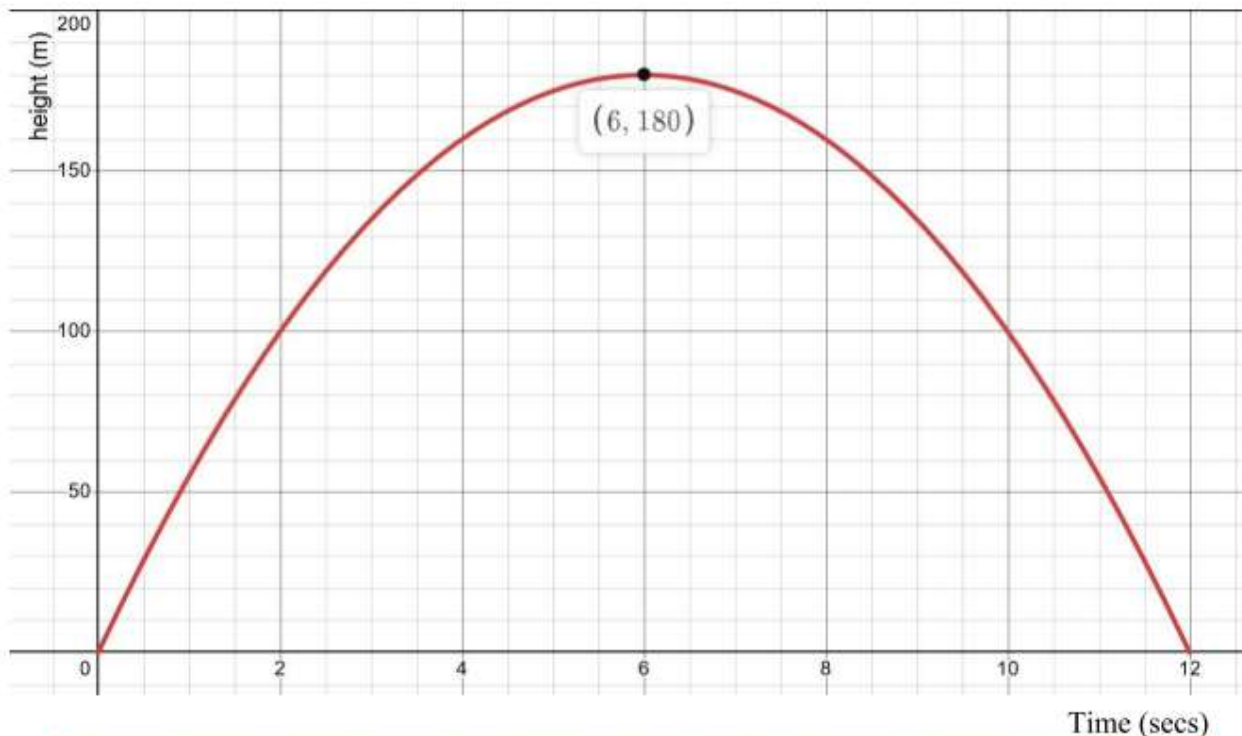
- find volume of pyramids, cones, spheres, and composite shapes, which includes prisms includes approximating, which could include through calculation, the volume of 3D objects. Prisms include cylinders and consist of more than rectangular prisms.

- (c) The path of a firework is shown in the graph below.

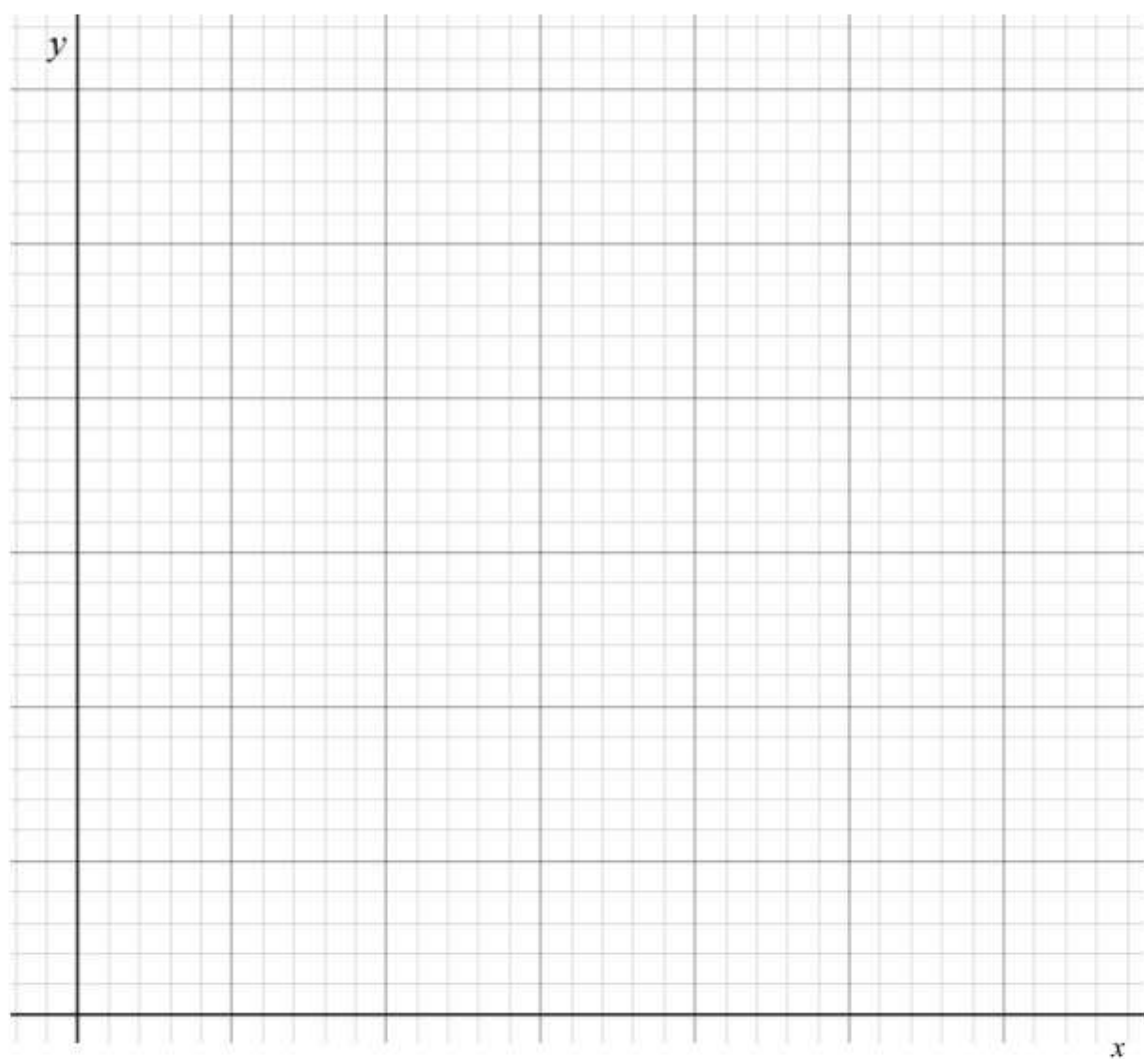
Its maximum point is at (6, 180).

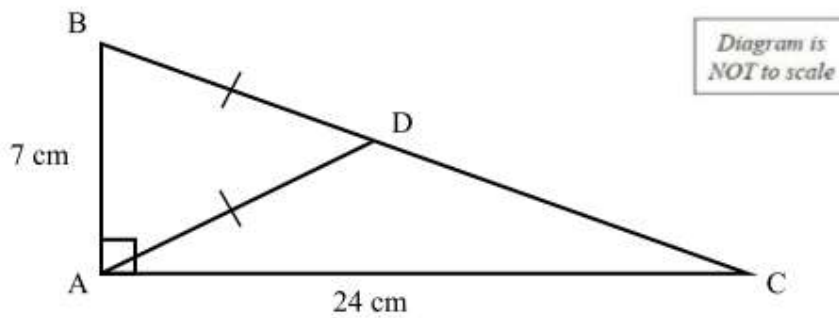
It lands 12 seconds after it was launched.

For how many seconds is the firework higher than 100 metres?



- graphing linear, quadratic, and exponential functions from patterns, tables, and equations, for example, making connections between representations such as number patterns, spatial patterns, tables, equations, and graphs.
- interpreting features of linear, quadratic, and exponential graphs in relation to the equation or the situation including x and y intercepts, gradient, vertex, asymptote, symmetry
- making links between different representations for the same model, for example, connect an equation, table and graph for distance-time graphs, filling a bath, rollercoaster ride, growth of organisms, compound interest





QUESTION TWO

- (a) \$8500 is invested for 5 years at 0.3% interest per year compound interest.

Given that the formula for compound interest is $Final\ Value = P\left(1 + \frac{r}{100}\right)^n$

where:

P = the initial amount invested

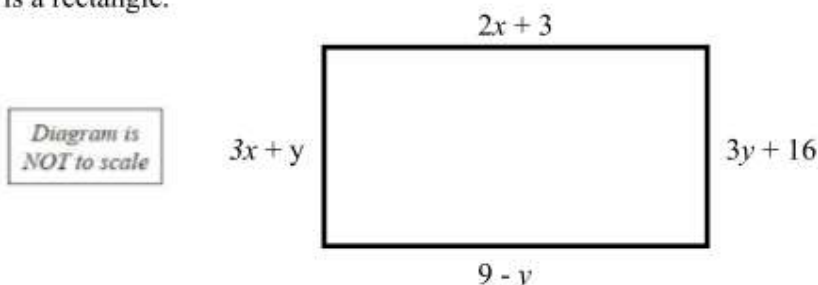
r = the interest rate

n = the number of years that the interest rate is applied

What is the final value of the investment after 5 years?

- rearranging, and using formulas (excluding the use of logarithms) – examples include: $d = \frac{1}{2}at^2$, $A = P(1 + nr)$, $t = 2\pi\sqrt{\frac{L}{g}}$
- making links between different representations for the same model, for example, connect an equation, table and graph for distance-time graphs, filling a bath, rollercoaster ride, growth of organisms, compound interest

- (b) The shape shown is a rectangle.



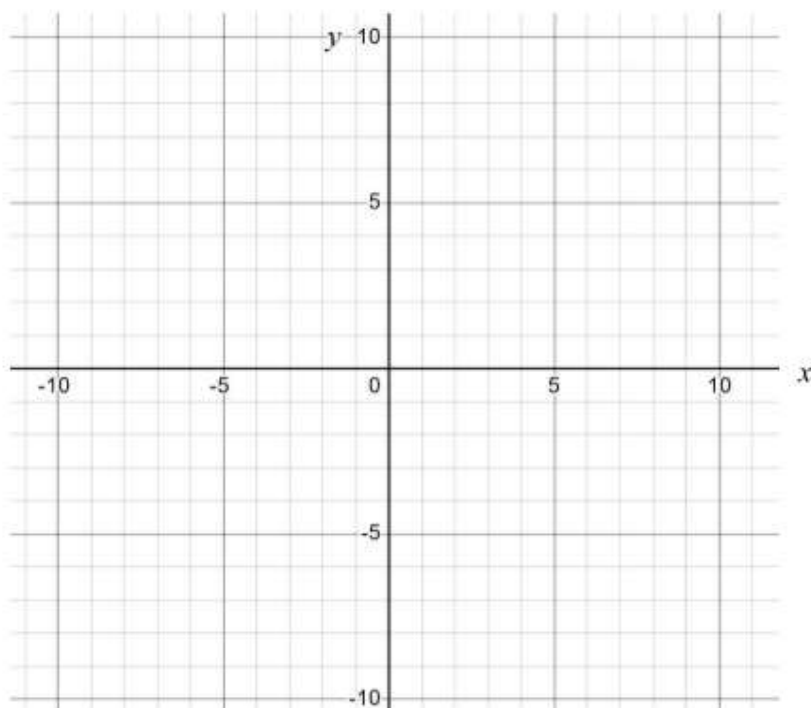
Form two equations and then find the values of x and y .

- use simultaneous linear equations with two unknowns, which includes solving given simultaneous equations, or forming and solving simultaneous equations. Methods for solving include substitution, elimination, graphical representation, graphing calculator and will not be prescribed. Note that solving linear with non-linear simultaneous equations are not included.

- (d) Points A, B and C have coordinates (2, 9), (10, -7) and (6, k) respectively.

Gradient of AB $\times$ gradient of AC = -1 (ie. They are perpendicular lines)

Find the value of k .



- interpreting features of linear, quadratic, and exponential graphs in relation to the equation or the situation including x and y intercepts, gradient, vertex, asymptote, symmetry

Note: $m_1 \times m_2 = -1$ means perpendicular gradients is out of scope for CL6

QUESTION THREE

- (a) Find the general rule for the n th term of the sequence:

5, 18, 35, 56, 81, 110,

- finding the equations for linear and quadratic functions, including horizontal and vertical lines, from patterns, tables or graphs

- (b) The shape ABCDV is a square-based pyramid.
O is the midpoint of the square base ABCD.

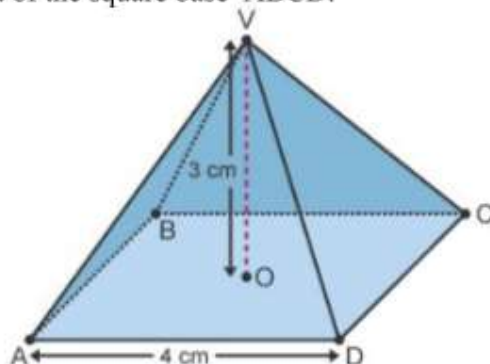


Diagram is
NOT to scale

Lengths AD, DC, BC and AB are all 4 cm.

The perpendicular height (OV) of the pyramid is 3 cm.

Calculate the angle between VC and the plane ABCD.

- use Pythagoras' theorem in right-angled triangles in 2D and 3D situations
- use trigonometric ratios in right-angled triangles in 2D and 3D situations

- (c) A 20cm length of wire is cut into two pieces, each of which is bent into a square. The length of the side of one square is x cm.

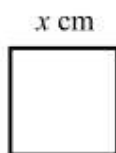


Diagram is
NOT to scale

The total area of the two squares is 14.5 cm^2 .

Find the lengths of the **two pieces of wire**.

- use linear and quadratic equations, which includes solving linear or quadratic equations, or forming and solving linear and quadratic equations from statements or drawings. Quadratic equations will have either $a = 1$ or be written in such a way that factorising is accessible for students without a graphing calculator.

Note: The quadratic created in this question is now deemed to be too difficult for students to factorise and solve as $a = 4$.

(d) Solve $\frac{3x^2-19x-14}{x^2-11x+28} = x + 5 - \frac{2x+1}{2}$

Assessment Schedule – 2023
Mathematics and Statistics: Demonstrate mathematical reasoning (91947)
Evidence Statement

Q1	Expected Coverage	Achievement (u)	Merit (r)	Excellence (t)
(a) $\frac{s}{t} = \left(\frac{u+v}{2} \right)$ $\frac{2s}{t} = u+v$ $u = \frac{2s}{t} - v$ Accept any equivalent arrangement		Correct rearrangement given.		
(b) $k = 2, k^3 = 8$ Volume of entire cone = $\frac{1}{3} \times \pi \times 15^2 \times 40$ $= 9424.78 \text{ cm}^3$ Smaller cone volume = $\frac{1}{8} \times 9424.78$ $= 1178.09 \text{ cm}^3$ Volume of frustrum = 8246.7 cm^3 (or equivalent)	Calculates volume of entire cone or Consistently calculates the volume of the smaller cone or frustrum	Correct volume of frustrum stated		
(c) Equation of parabola: $y = kx(x - 12)$ $180 = k(6)(6 - 12)$ $k = -5$ $\therefore y = -5x(x - 12)$ or equivalent Time: $-5x^2 + 60x = 100$ $-5x^2 + 60x - 100 = 0$ $x^2 - 12x + 20 = 0$ $(x - 2)(x - 10) = 0$ $x = 2, 10$ The firework is higher than 100m for 8 seconds	A time of 8 seconds found from reading the values from the graph or Correct equation of parabola found	Quadratic consistently factorised and solved	Correct solutions stated and	The length of time that the firework is above 100m stated

(d)	$BC^2 = \sqrt{7^2 + 24^2}$ $BC = 25$ $\tan B = \frac{24}{7}$ $B = 73.7^\circ$ $\cos 73.7 = \frac{3.5}{BD}$ $BD = 12.5 \text{ cm}$ $\therefore CD = 25 - 12.5$ $CD = 12.5 \text{ cm}$			Length of BC found using Pythagoras' Theorem and Either angle B or angle C found using trigonometry	Consistent length found after minor error made in Pythagoras or trigonometry	Correct length found using valid methods
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N0	N1	N2	A3	A4	M5	M6	E7	E8
No response	1 of N1	2 of N1	1 of u	2 of u	1 of r	2 of r	1 of t	2 of t

Note: N1 is an indication as some correct work done in part of a question but not enough to get a u grade – these are judgement calls.

Q2	Expected Coverage	Achievement (u)	Merit (r)	Excellence (t)
(a)	$8500 \left(1 + \frac{0.3}{100}\right)^5 = \8628.27	Correct value stated		
(b)	$2x + 3 = 9 - y$ (1) $3x + y = 3y + 16$ (2) $(1) y = -2x + 6$ $(2) 3x - 2y = 16$ Subst (1) into (2) $3x - 2(-2x + 6) = 16$ $7x = 28$ $x = 4$ $y = -2$	Both equations formed and the process of substitution or elimination started or Both equations formed and correct solution stated (from GC)	Both x and y found through a process of elimination or substitution	
(c)	$\frac{2\pi rh + 2\pi r^2}{4\pi r^2} = 2$ $2\pi rh + 2\pi r^2 = 8\pi r^2$ $h + r = 4r$ $h = 3r$ $\frac{\pi r^2(3r)}{\frac{4}{3}\pi r^3} = \frac{9}{4}$	The final fraction is consistently found from an incorrect value for h	Correct value for h found	Correct value for h found and correct formulas substituted into fraction and The final fraction is correctly found

(d)	$AB: y = -\frac{16}{8}x + c$ $9 = -2(2) + c$ $c = 13$ $y = -2x + 13$ $AC: y = \frac{1}{2}x + c$ $9 = \frac{1}{2}(2) + c$ $c = 8$ $y = \frac{1}{2}x + 8$ $y = \frac{1}{2}(6) + 8$ $y = 11$ So $k = 11$			Equation for the line AB found or Consistent equation for line AC found	Equations for line AB AND line AC found	Equations for lines AB and AC found and The value of k stated
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N0	N1	N2	A3	A4	M5	M6	E7	E8
No response	1 of N1	2 of N1	1 of u	2 of u	1 of r	2 of r	1 of t	2 of t

Q3	Expected Coverage	Achievement (u)	Merit (r)	Excellence (t)
(a)	$y = (2n - 1)(n + 4)$ $y = 2n^2 + 7n - 4$ or equivalent	Correct quadratic stated (accepted factorised)		
(b)	$AC^2 = 4^2 + 4^2$ $AC = \sqrt{32}$ $OC = 2.83 \text{ Cm}$ $\tan(VCO) = \frac{3}{2.83}$ $\text{Angle } VCO = 46.7^\circ$	Length of AC correctly found or Angle VCO consistently found	Angle VCO correctly found	
(c)	Second square has side length $5 - x$ $x^2 + (5 - x)^2 = 14.5$ $2x^2 - 10x + 25 = 14.5$ $4x^2 - 20x + 21 = 0$ $(2x - 3)(2x - 7) = 0$ $x = \frac{3}{2}, \frac{7}{2}$ First length of wire = 6 cm Second length of wire = 14 cm	Recognising that the side length of second square is $5 - x$ or Consistently solving a quadratic to give 2 solutions	Correct quadratic formed and rearranged to make it equal to zero or Consistent lengths stated from a minor error prior	Correct solutions found and correct lengths stated for each square
(d)	$\frac{(x - 7)(3x + 2)}{(x - 7)(x - 4)} = x + 5 - \left(\frac{2x + 1}{2}\right)$ $\frac{3x + 2}{2x + 10 - 2x - 1} = \frac{x - 4}{6x + 4} = 9$ $\frac{x - 4}{6x + 4} = 9$ $6x + 4 = 9x - 36$ $3x = 40$ $x = \frac{40}{3}$ or equivalent	Factorised numerator on LHS or Cross multiplies and simplifies numerator to get 2 as denominator of RHS	Simplifies to $\frac{6x+4}{x-4} = 9$ or equivalent	Correct solution stated

N0	N1	N2	A3	A4	M5	M6	E7	E8
No response	1 of N1	2 of N1	1 of u	2 of u	1 of r	2 of r	1 of t	2 of t

Cut scores

	Not Achieved	Achievement	Achievement with Merit	Achievement with Excellence
Score range	0 – 6	7 – 12	13 – 18	19 – 24